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Aftertreatment Selective Catalytic Reduction (SCR) Reuse Guidelines

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I. Introduction

- This document provides reuse guidelines for the aftertreatment selective catalytic reduction (SCR), in conjunction with the information provided within the Service Procedures listed in Table 7 below.
- The reuse guidelines contained in this document are the **only** approved inspection methods and pass/fail criteria for inspecting aftertreatment SCR for reuse.

II. Reuse Guidelines Summary

Aftertreatment Selective Catalytic Reduction (SCR) Reuse Guidelines

Component	Corresponding Table and Figure	Condition	Approach	Additional Remarks
SCR Inlet	1	New with writing on face	Reuse	See Table 1 below
SCR Inlet	2	Iron oxide deposits	Reuse	See Table 2 below
SCR Inlet	3	DEF deposits	Conditional Reuse	See Table 3 below
SCR Inlet	4	Erosion	Reuse	See Table 4 below
SCR Inlet	5	Water stains	Conditional Reuse	See Table 5 below
SCR Inlet	6	Sodium deposits	Conditional Reuse	See Table 6 below

Aftertreatment SCR Housing Reuse Guidelines:

- The aftertreatment SCR housing **must** be free of dents.
- All sealing flanges **must** be free of dents, cracks, or gouges to seal properly with gaskets.

III. Pictures of New Components

New Aftertreatment SCR



Figure 1, New Aftertreatment SCR.

Table 1, New Aftertreatment SCR

Appearance	Clean with no dents or gouges and writing on face.
Comment	Ready for use.

IV. Failure Modes and Reuse Pictures

Aftertreatment SCR with Iron Oxide Deposits



Figure 2, Aftertreatment SCR Inlet with Iron Oxide Deposits.

Table 2, Aftertreatment SCR – Iron Oxide Deposits

Appearance	Brown discoloration on aftertreatment SCR inlet face.
Cause	Normal engine operation.
Action	None, can continue to be used.

Aftertreatment SCR Inlet Face with Diesel Exhaust Fluid (DEF) Deposits

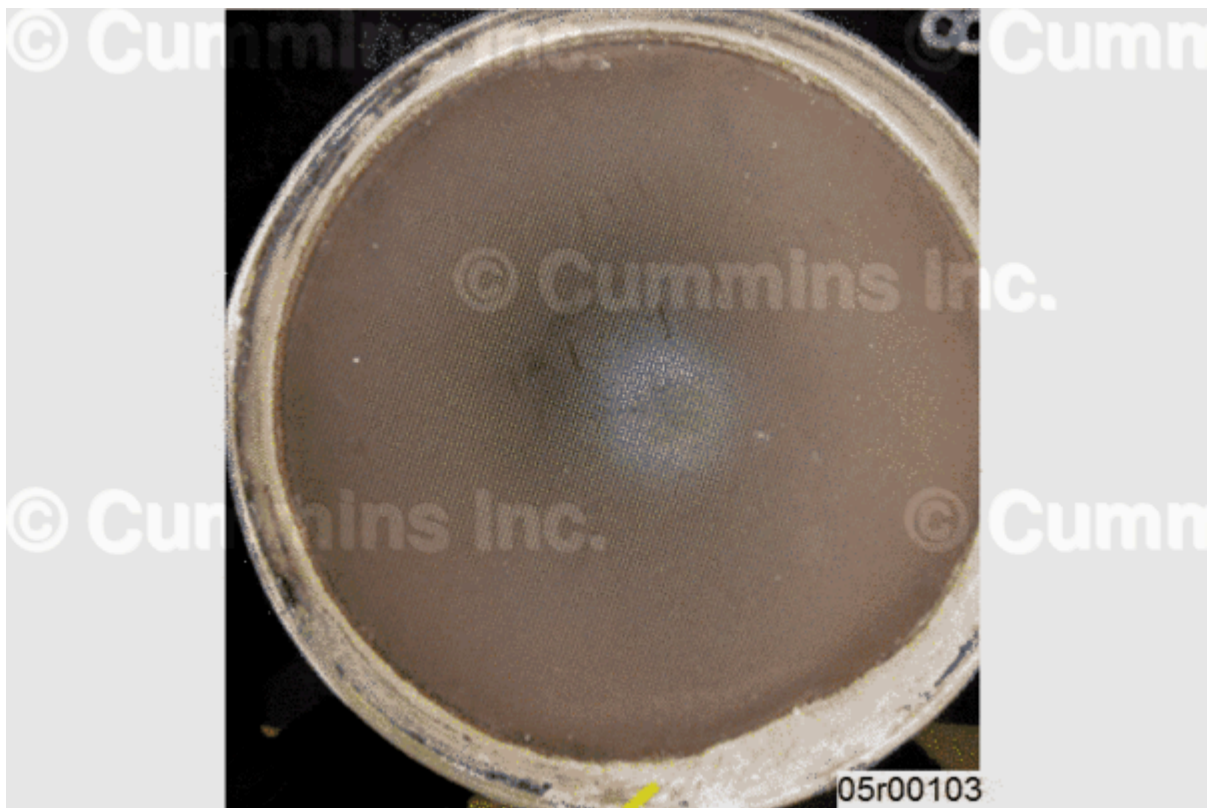


Figure 3, Aftertreatment SCR Inlet with DEF Deposits.

Table 3, Aftertreatment SCR – DEF Deposits

Appearance	White material build up on outer edge of aftertreatment SCR inlet face.
Cause	Normal engine operation.
Action	Clean aftertreatment SCR. See corresponding Service Manual. Reference Procedure 011-036 in Section 11.

Aftertreatment SCR Inlet Face Media Erosion



Figure 4, Aftertreatment SCR Inlet Media Erosion.

Table 4, Aftertreatment SCR – Media Erosion

Appearance	Uneven wear on aftertreatment SCR inlet face.
Cause	Normal engine operation .
Action	None, can continue to be used.

Aftertreatment SCR Inlet Face with Water Stains

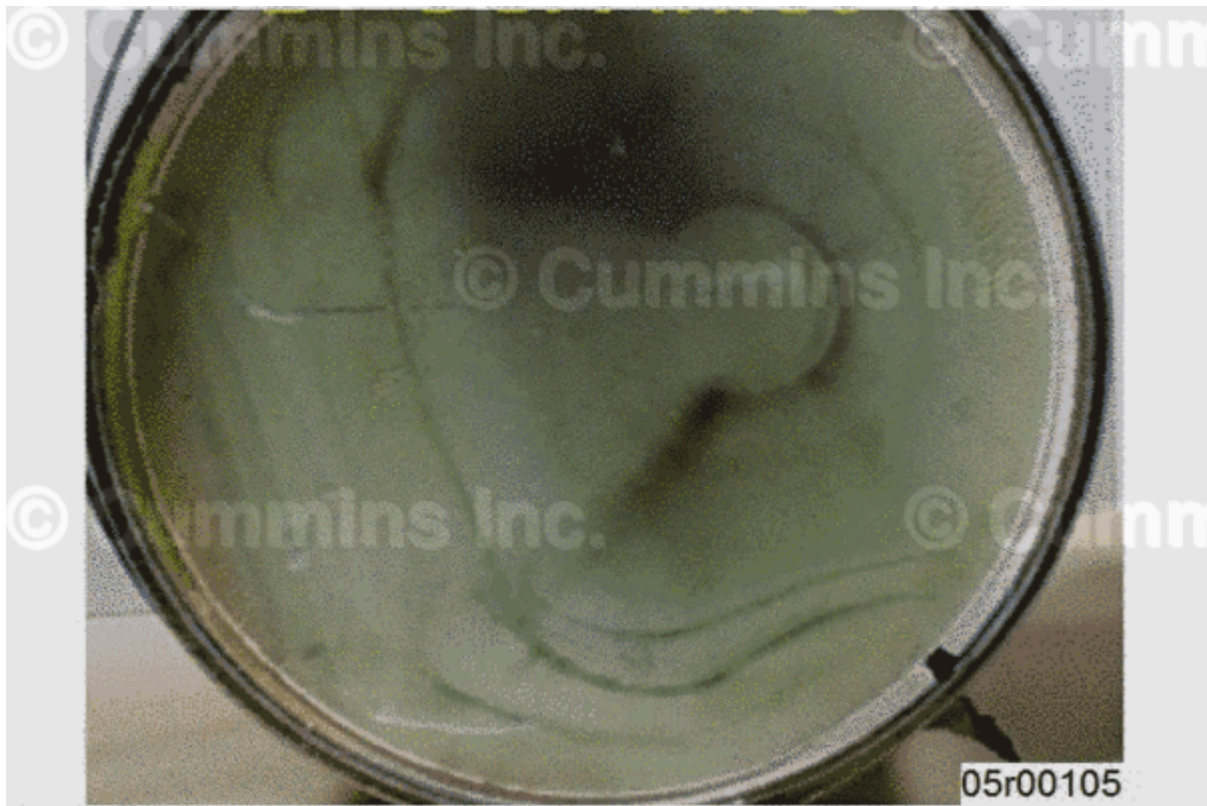


Figure 5, Aftertreatment SCR Inlet with Water Stains.

Table 5, Aftertreatment Selective Catalytic Reduction (SCR) Inlet – Water Stains

Appearance	Patterns of green and dark stains on aftertreatment SCR inlet face.
Cause	Moisture in exhaust system.
Action	Check aftertreatment SCR drain holes for plugging. Check for aftertreatment SCR conversion efficiency fault codes. If fault codes are present, follow applicable troubleshooting trees to address fault codes.

Aftertreatment SCR Inlet with Soot Build Up.

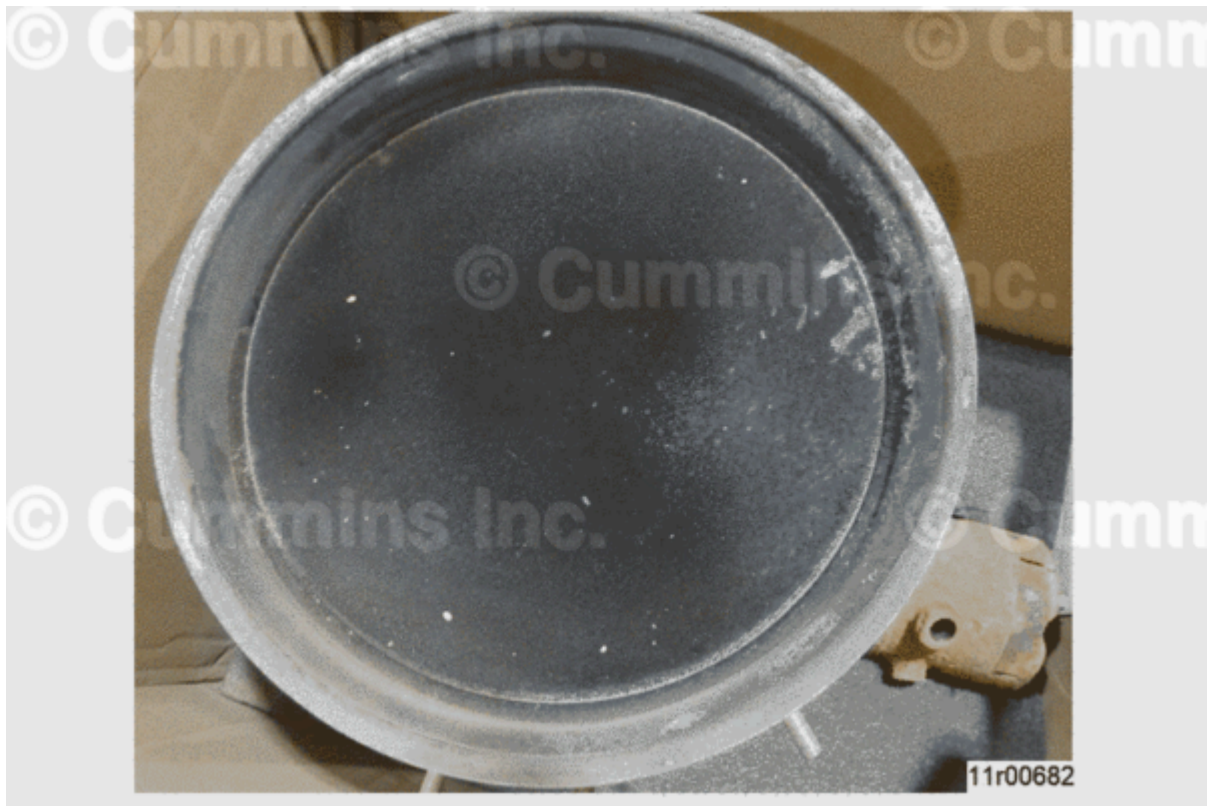


Figure 6, Aftertreatment SCR Inlet Face with Soot Stains.

Table 6, Aftertreatment Selective Catalytic Reduction (SCR) – Soot Stains

Appearance	Soot stains on aftertreatment SCR inlet face.
Cause	Malfunctioning aftertreatment DPF.
Action	Check aftertreatment SCR drain holes for plugging. Perform a stationary regeneration. Check for aftertreatment SCR conversion efficiency fault codes. If fault codes are present, follow applicable troubleshooting trees to address fault codes.

Aftertreatment SCR Inlet with Sodium Deposits

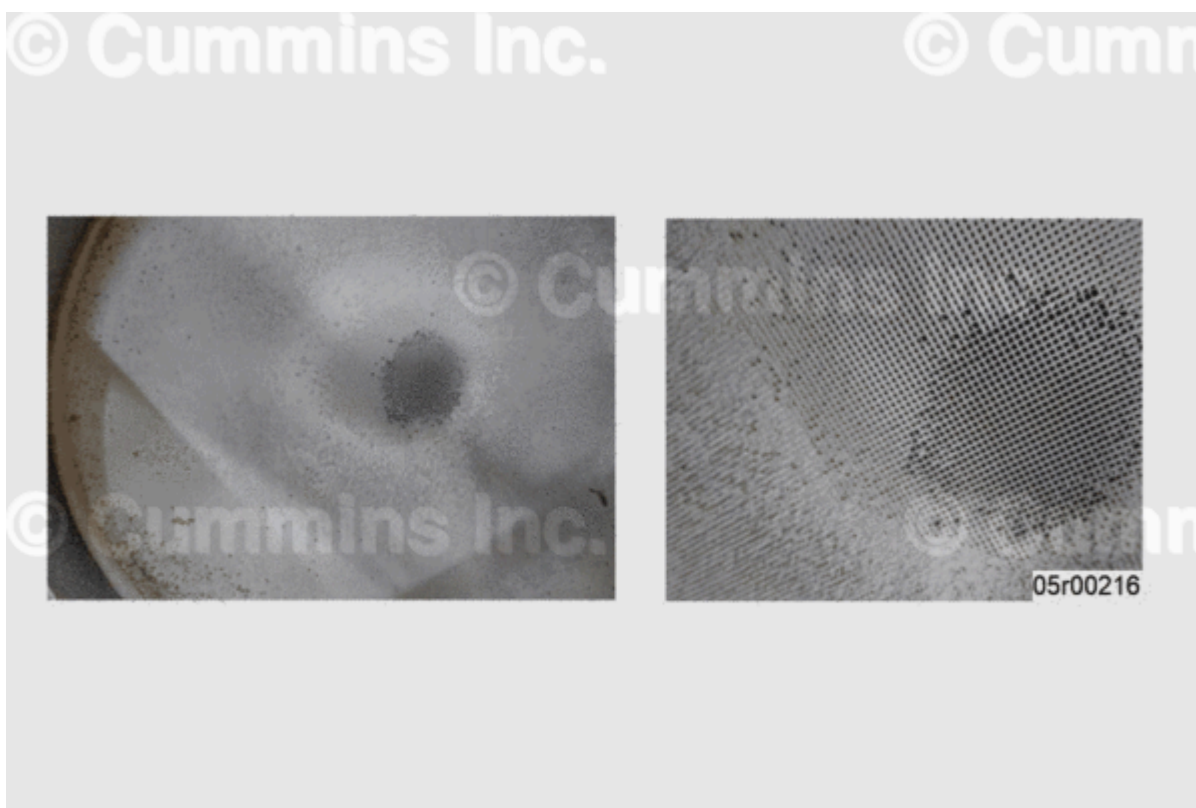


Figure 7, Aftertreatment SCR Inlet with Sodium Deposits.

Table 7, Aftertreatment Selective Catalytic Reduction (SCR) Inlet – Sodium Deposits

Appearance	White or Brown material build up on aftertreatment SCR inlet face.
Cause	High Sodium level in DEF caused by water dilution or incorrect/non-approved DEF
Action	Verify a stationary regeneration has been performed. If DPF out pressure high fault codes persist, the SCR catalyst must be replaced.

V. Inspection

An aftertreatment SCR can show damage through signs of contamination with coolant, oil, or fuel.

Repair Direction: Any oil, fuel, or coolant-soaked aftertreatment SCR catalyst **must** be replaced.

VI. Associated Procedures

Table 8, Associated Procedures

Manual Type	Procedure Title	Procedure	Section
Service Manual	Aftertreatment Selective Catalytic Reduction (SCR)	011-036	11

Document History

Date	Details
2017-5-19	Module Created
2020-8-25	Reformatted sections. Added Figure 6 and Table 6 to Section IV.
2023-3-2	Added Aftertreatment SCR Inlet Face with Soot Stains.

Last Modified: 09-Mar-2023
